



A novel generalized source-free domain adaptation approach for cross-domain industrial fault diagnosis

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ABSTRACT

Domain adaptation has been widely applied in data-driven fault diagnosis tasks to address the domain shift problem between source and target data. However, conventional domain adaptation methods require both domains to be known, which is not always feasible due to privacy concerns and big data transmission. To overcome this limitation, a dedicated method called source-free domain adaptation (SFDA) has been developed to ensure reliable performance without relying on source data during target model adaptation. SFDA can achieve accurate classification tasks under domain shift problems and source data-free scenarios. We propose a generalized source model with manifold Mixup data augmentation and label smoothing techniques to avoid overfitting during the source model training. Based on this model, a novel self-training framework is proposed to implement the domain adaptation task and achieve accurate prediction performance. The experimental results from three real-world datasets demonstrate the effectiveness of the proposed approach.

1. Introduction

1.1. Background and related works

As Industry 4.0 continues to evolve, industrial systems are increasingly vital in all production processes [1]. Industrial systems will be gradually aging and fail under the conditions of prolonged usage and the harsh environment [2]. Without corresponding maintenance measures, there is a significant threat to production safety and reliability [3]. As a result, fault diagnosis and health management have received considerable attention in recent years from academia and industry [4]. The essential technique requires a combination of human knowledge and experience to perform reasonable health monitoring and complete accurate fault diagnosis of the system, which helps in timely maintenance [5].

Model-based approaches, combining human prior knowledge and physical model generation, have been widely used in fault diagnosis. However, these related models struggle to adapt to increasingly complex and integrated industrial devices, which cannot achieve accurate fault diagnosis tasks. Unlike such physical model-based approaches, intelligent fault diagnosis can reduce the dependence on prior physical knowledge, which has recently been developing rapidly [6]. Intelligent data-driven approaches can continuously learn large amounts of existing healthy and faulty data through model learning, which can

accurately identify faulty classes without physical knowledge [7]. Conventional machine learning methods are commonly used for intelligent fault diagnosis tasks, such as support vector machines (SVM) [8], decision trees (DT) [9], random forest (RF) [10], multi-layer perceptive (MLP), etc. Such models can implement fault diagnosis tasks in many applications.

Due to the limitations of the model design, its weak nonlinear processing capability is difficult to adapt to complex and integrated industrial systems, and cannot solve complex fault diagnosis problems. Therefore, deep learning-based methods can effectively address this problem due to their powerful nonlinear processing capabilities [11]. Thanks to the continuous development of computer theory and technology, trained neural networks models based on forward propagation and gradient descent can well solve various encountered problems and hence achieve accurate diagnostic tasks. Such an end-to-end pattern recognition approach can construct complex and feasible nonlinear mappings from sensor data to faulty classes by training on a large amount of data [12]. There are lots of researchers that have studied the task of fault diagnosis based on deep learning. Zhang et al. proposed an intelligent methods for fault diagnosis task with denoising Autoencoder and self-attention mechanism [13]. Ye et al. proposed an auto-encoder model with metric learning for fault diagnosis task [14]. According to the results of these papers, the deep learning based methods can also achieve the better diagnostic performance.

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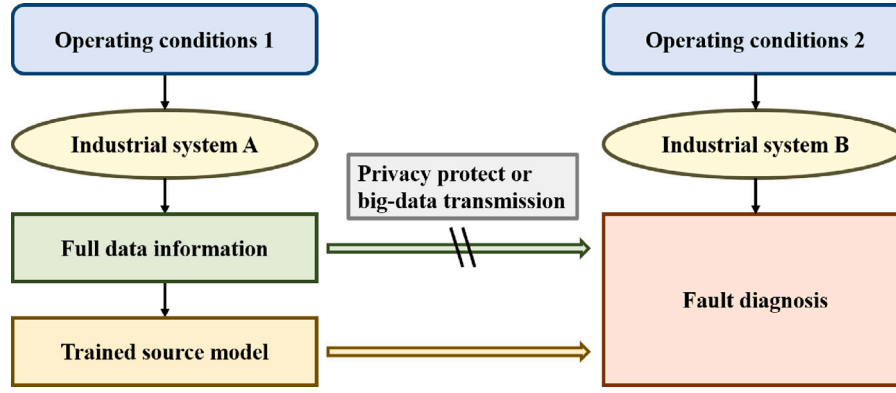


Fig. 1. The motivation of source-free domain adaptation methods on cross-domain fault diagnosis.

However, such deep learning based methods have their own shortcomings in terms of data quality. It has a very important premise that the training and test datasets must meet the independent identical distribution (IID) assumption [15]. Due to the complex environment and the presence of unknown influences within the device, it is difficult to collect data with the same distribution in actual situations and hold the hypothesis in practice [16]. Therefore, existing distributions of training and test data tend to be different. Such non-IID conditions would break the premise of the algorithm and reduce the generalization of the models, increasing the upper bound of its prediction error. To tackle the problem, domain adaptation techniques have been developed to reduce the distribution discrepancy and continuously shorten the distance by setting an optimization objective [17]. Compared with the conventional method, the adaptation loss is added, as shown in Fig. 2(a). Due to the type of optimization objective, domain adaptation techniques can be categorized into metric-based and adversarial-based [15]. Metrics-based methods rely on statistical metrics, such as Maximum Mean Discrepancy (MMD) [18], Correlation Alignment (Coral) [19]. The optimization object is to reduce the metric during training and obtain domain-invariant features that are strongly correlated with the class and weakly correlated with other factors such as the environment and noise [20]. Adversarial-based methods can modify the metric into a new domain classifier, which can help achieve domain-invariant features in both the source and target domains. Chai et al. developed a more detailed adversarial model for diagnosing the industrial faults facing cross-domain conditions, and achieve a high-precision performance [21]. Tan et al. proposed a Mixup-based domain adaptation model for the fault diagnosis task to address the domain shift problem [22].

The above methods can effectively improve the domain shift problem and have theoretical support. However, the source data is unavailable for the target model adaptation process due to the challenges for privacy protection and big data transmission issues [23]. For important industrial processes, source domain data is often unavailable and requires strict confidentiality due to confidential information as shown in Fig. 1. Meanwhile, large scale of source data and its related computation burden can also make it unpractical especially in real-time scenario. Therefore, the above effective methods cannot be applied under source-free conditions [24]. It is of great practical importance to design such a source-free domain adaptation, and the difference can be shown in Fig. 2. In this case, the source model generation needs to ensure its generalization ability to prevent over-fitting, which is beneficial for the target fault diagnosis task. At the same time, it is necessary to find a method for target model adaptation that can still achieve domain adaptation even with the unknown source data. Therefore, there are ample motivations to address these issues in this paper as follows.

- Intelligent fault diagnosis is essential for the prognostics and health management, which can provide accurate maintenance

decision information and reduce related economic losses while reducing system mechanism dependence through data-driven methods.

- In order to improve the performance degradation caused by domain shift, it is crucial to make sure that the trained model in the source domain has strong generalization ability.
- There have been few studies on domain adaptation fault diagnosis problems with unavailable source data. How to maximize the use of target-domain data and source-domain models to achieve domain adaptation and reduce domain shift is an urgent need for research.

1.2. Research and contributions of proposed works

In this sub-section, we present the related works addressing the challenging and meaningful issues of source-free domain adaptation. The training task of SFDA tasks is modified as shown in Fig. 2(b), which includes source model generation and target domain adaptation [23]. Compared to Fig. 2(a), these two processes are no longer carried out simultaneously due to the limitations of unknown source data. Target domain adaptation is a crucial and typical unsupervised learning step based on a known source model under the source data-free scenarios. To tackle the problem, self-training methods with pseudo-labels and iterations are essential techniques [25]. The main objective of these methods is to leverage existing knowledge to generate and iterate pseudo-labels while using some noise reduction methods to reduce output uncertainty [26]. Recently, a few target domain adaptation methods for fault diagnosis tasks have been proposed. Jiao et al. presented a nuclear norm-based target domain adaptation approach [27]. Li et al. proposed a confidence-filter-based domain adaptation framework to enhance the trustworthiness of pseudo labels [28].

The existing SFDA works have not extensively addressed the issue of overfitting caused by domain shift, which can reduce the effectiveness of unsupervised domain adaptation. To overcome this limitation, a generalized source model is required that avoids overfitting, reduces the model's overconfidence in training samples (source), and improves generalization to the target. Some studies have focused on enhancing model generalization, such as label smoothing [29] and Mixup [30]. Label smoothing improves the traditional cross-entropy loss by avoiding overconfidence on the current sample and overfitting caused by one-hot encoding [29]. Mixup is a data augmentation method that addresses overfitting problems by treating the instance-space as a continuous distribution, improving smoothness within the neighborhood of samples [31].

Based on the above motivations and current research, we combine a manifold Mixup module and label smoothing technique into a source data-free framework to realize more effective and generalized fault classification tasks. And our contributions can be summarized as follows:

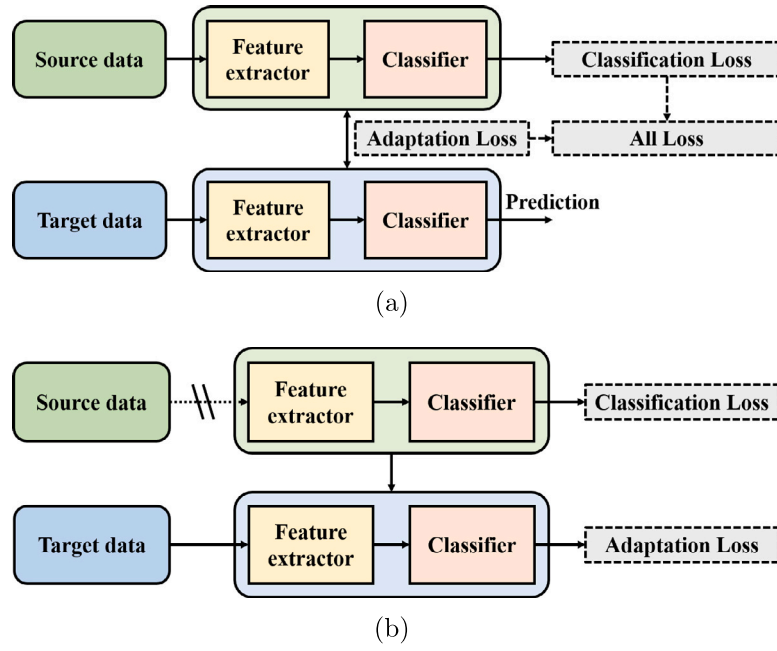


Fig. 2. Diagram of conventional domain adaptation (Top) and source-free domain adaptation methods (Bottom). The main difference is whether the source data is available during target model adaptation.

- A novel source-free domain adaptation approach is proposed to enable fault diagnosis in industrial systems, eliminating dependence on source data and reducing traditional domain adaptation's resource consumption. During target model adaptation, a self-training strategy is applied with pseudo-labels and uncertainty reduction.
- During the source model training process, a deep generalized neural network is proposed with a well-designed manifold Mixup module internal to the model and trained jointly. We improve the generalization of the source model to benefit the model adaptation.
- We validate the proposed model using three real-world datasets and obtain better performance compared to conventional and state-of-the-art methods.

The remainder of the paper can be organized as follows. Section 2 formulates the proposed methodology and algorithm implementation in detail. Section 3, 4 and 5, present the experimental results on Case 1, 2, and 3 respectively. Finally, Section 6 concludes the paper.

2. Proposed source-free domain adaptation method

2.1. Overview

Two domains, including source and target datasets, can be considered in cross-domain conditions to realize the fault diagnosis task. Source domain D_S contains the inputs and fault classes with data couples $\{x_S, y_S\}$, where $x_S \in \mathcal{X}_S$ and $y_S \in \mathcal{Y}_S$. And target domain D_T contains $\{x_T, y_T\}$, where $x_T \in \mathcal{X}_T$ and $y_T \in \mathcal{Y}_T$. As a closed-set domain adaptation method, there is an assumption that the source and target domains share the same label space, where $\mathcal{Y}_S = \mathcal{Y}_T$. The conventional domain adaptation of intelligent fault diagnosis task is to learn a mapping function $f_T : x_T \rightarrow y_T$ from known $\{x_S, y_S\}$ and $\{x_T\}$. In SFDA task [32], the known condition is constrained to trained mapping f_S and known $\{x_T\}$.

In vanilla SFDA tasks [33], there are two main stages for domain adaptation, including the training of a generalized model for the source datasets and the self-training for the target domain adaptation. The first stage is used to obtain the mapping function f_S from the source data.

The second stage of model adaptation is achieved by self-training fine-tuning, which is a semi-supervised learning process based on the source model.

2.2. Generalized model training under source domains

The diagram of proposed generalized model training for source domains is shown in Fig. 3. For the deep neural network of the diagnostic model, it consists of two modules: a feature extractor g and a fault classifier h .

The feature extractor module g can be used to build a mapping $G_g : \mathcal{X} \rightarrow \mathbb{R}^g$, which can obtain the transferable features based on the domain adaptation task. Such domain-invariant information can reduce the negative impact of domain shift. The forward calculation of function $G_g(x_S)$ is shown in Eq. (1):

$$g_S = G_g(x_S) = \sigma(W_k^g(\dots \sigma(W_1^g x_S + b_1^g) \dots) + b_k^g) \quad (1)$$

where W_k^g and b_k^g mean the weights and bias of the k th layer of function $G_g(\cdot)$. x_S is the data input of source, g_S is the domain-invariant feature, and σ is the activation function. In this paper, we proposed the Rectified Linear Unit (ReLU).

The fault classifier module is trained behind the feature extractor to build a mapping $G_h : \mathbb{R}^g \rightarrow \mathbb{R}^C$, and the detailed format is shown below:

$$\begin{aligned} q_{pred} &= G_h(g_S) = S(\underbrace{W_k^h(\dots \sigma(W_1^h g_S + b_1^h) \dots) + b_k^h}_{\Theta_{h,c}(g_S)}) \\ &= \left[\frac{\exp(\Theta_{h,c}(g_S))}{\sum_{j=1}^C \exp(\Theta_{h,j}(g_S))} \right]_{c=1}^C \end{aligned} \quad (2)$$

where W_k^h and b_k^h mean the weights and bias of the k th layer of function $G_h(\cdot)$. $\Theta_{h,c}$ is the mapping from domain-invariant features to the corresponding c th output. S is the softmax activation function, which outputs $\{0, 1, \dots, C\}$ by using one-hot encoding mode. q_{pred} is also the one-of- C encoding.

Based on the known source data $\{x_S, y_S\}$, the source mapping function f_S can be trained by the conventional supervised learning strategy.

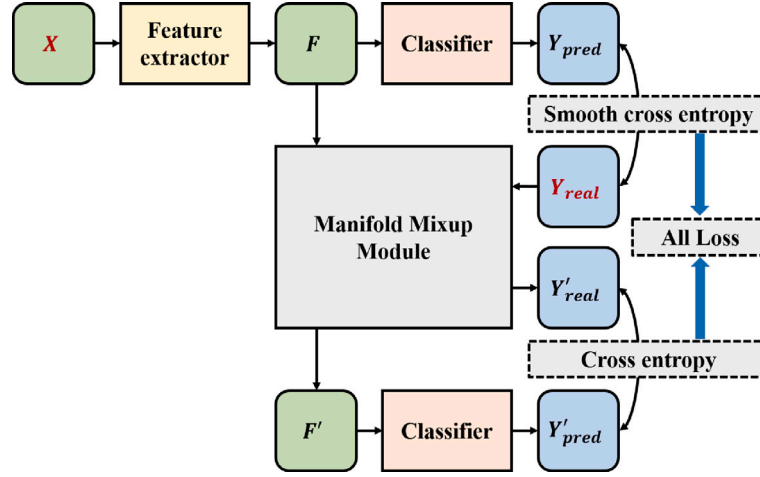


Fig. 3. Diagram of generalized source model generation with label smoothing and manifold Mixup data augmentation.

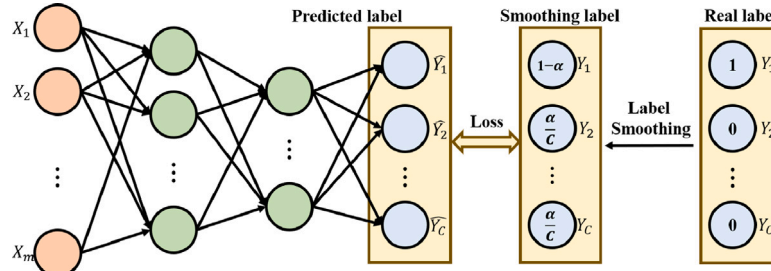


Fig. 4. Detailed diagram of label smoothing.

To improve the generalization of classifier, the label smoothing cross entropy is used to as a main optimization objective, as shown below:

$$L_S^{(1)} = -\mathbb{E}_S \left(\sum_{i=1}^C \hat{q}_i \log \frac{\exp(\Theta_{h,i}(g_S))}{\sum_{j=1}^C \exp(\Theta_{h,j}(g_S))} \right) \quad (3)$$

where $\mathbb{E}_S(\cdot)$ means the mean outputs of the whole batch-size source samples. q_i is the one-of-K encoding of the real y_S , where the correct class corresponds to a value of 1 and the rest classes correspond to a value of 0. $\hat{q}_i = (1-\alpha)q_i + \alpha/C$ means the smoothing labels, α is the smoothing parameter. The diagram of label smoothing is shown in Fig. 4. To improve the fault classifier's capability of generalization, we utilize a manifold Mixup module. It serves as a reliable data augmentation technique, which can generate new domain-invariant feature-level pairs [34]. Let g_i and g_j are feature extractor outputs of the i th and j th samples. q_i and q_j are the corresponding classes with one-of-C encoding format. A new feature pairs can be calculated as:

$$g'_i = \text{Mix}_\lambda^g(g_i, g_j) = \lambda g_i + (1-\lambda)g_j \quad (4)$$

$$q'_i = \text{Mix}_\lambda^q(q_i, q_j) = \lambda q_i + (1-\lambda)q_j \quad (5)$$

where Mix_λ^g and Mix_λ^q are the manifold Mixup mappings from (g_i, g_j) to g'_i and from (q_i, q_j) to q'_i . λ is a random variable, ranging from 0 to 1, and sampled from beta distribution $\text{Beta}(\beta, \beta)$ for $\beta \in (0, +\infty)$. β is a hyper-parameter of manifold Mixup data augmentation, which is equal to 0.2 [34].

The manifold Mixup module extends the data distribution by linear interpolating a combination of features f from the known training set, as shown in Fig. 5. It can play a crucial role in enhancing domain generalization problems through powerful data augmentation. Based on

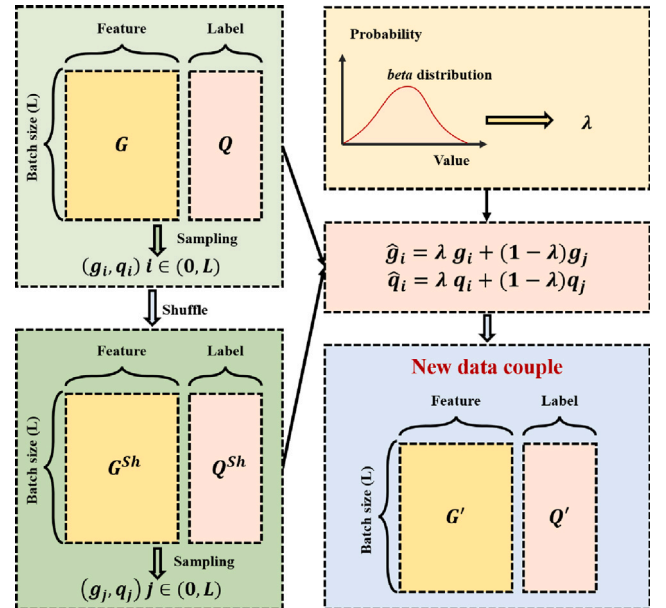


Fig. 5. Diagram of manifold Mixup data augmentation module on the source training stage.

the manifold Mixup augmentation, the new data pairs can be obtained as follows:

$$\{g'_i, q'_i\} = \{\text{Mix}_\lambda^g(g_i, g_j), \text{Mix}_\lambda^q(q_i, q_j)\} \quad (6)$$

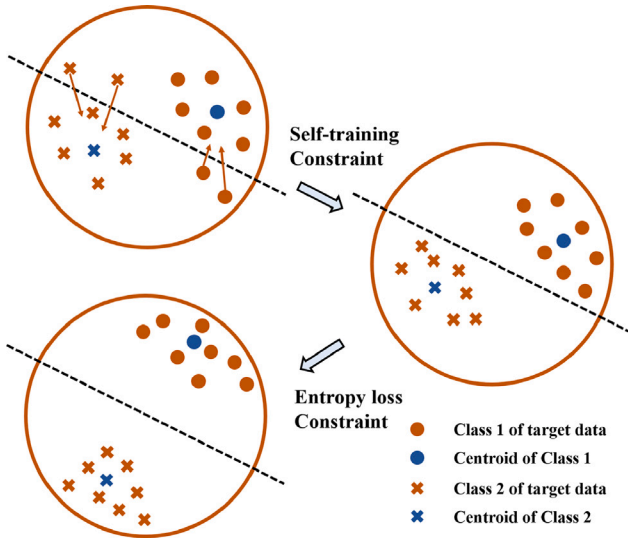


Fig. 6. Diagram of proposed target domain adaptation (including the self-training pseudo-labels and self-entropy loss constraint).

To calculate the predicted faulty classes, the fault classifier can proceed as follows:

$$q'_{pred} = G_h(g'_S) = S(W_k^h(\dots \sigma(W_1^h g'_S + b_1^h) \dots) + b_k^h) \quad (7)$$

$$= \left[\frac{\exp(\Theta_{h,c}(g'_S))}{\sum_{j=1}^C \exp(\Theta_{h,j}(g'_S))} \right]_{c=1}^C$$

where g'_S is the data generated by the manifold Mixup method, q'_{pred} means the prediction result with the one-hot encoding format. Therefore, we define the loss function $L_S^{(2)}$ for the insert manifold Mixup module as follows:

$$L_S^{(2)} = -\mathbb{E}_S \left(\sum_{i=1}^C q'_i \log q'_{pred} \right) \quad (8)$$

$$= -\mathbb{E}_S \left(\sum_{i=1}^C q'_i \log \frac{\exp(\Theta_{h,i}(g'_S))}{\sum_{j=1}^C \exp(\Theta_{h,j}(g'_S))} \right)$$

where $\mathbb{E}_S$ is the mean output of each batch-size source samples. q'_i is the related labels by manifold Mixup function Mix_q^g .

We integrate the label smoothing cross entropy $L_S^{(1)}$ of normal forward calculation and cross entropy $L_S^{(2)}$ of manifold Mixup module. Eq. (9) defines the ultimate loss function of source model training.

$$L_S^{all} = L_S^{(1)} + \lambda_1 L_S^{(2)} \quad (9)$$

where λ_1 is the balancing hyper-parameter. The gradient descent process of source model parameters is designed as shown as Eqs. (10) to (11):

$$\theta_g^t = \theta_g^{t-1} - \epsilon_S \cdot \frac{\partial L_S^{all}}{\partial \theta_g} \quad (10)$$

$$\theta_h^t = \theta_h^{t-1} - \epsilon_S \cdot \frac{\partial L_S^{all}}{\partial \theta_h} \quad (11)$$

where ϵ_S is the learning rate of training process. θ_g^t and θ_h^t contain the parameters including weights and bias of feature extractor and fault classifier at t epoch.

2.3. Self-training target domain adaptation

The source model can be generated with strong generalization by label smoothing cross-entropy and manifold Mixup data augmentation.

In contrast to conventional domain adaptation, the source data is unknown during target training, and other flexible adaptation methods are needed to update the model.

We propose two main optimization objectives for target model adaptation, including the self-training pseudo-label cross-entropy and self-entropy loss constraints. A detailed diagram of these two steps is shown in Fig. 6. Through the pseudo-label setting and successive iterations, the self-training loss constraint can be designed to reduce the samples near the classification interface and the number of misclassified samples due to domain shift. The associated loss functions are designed during the training of the target model.

2.3.1. Self-training constraint

After the source model generation process, the fault classifier has already a good pattern recognition ability towards faulty classes. There is no need to train the classifier in self-training learning to avoid negative transfer. Therefore, the same fault classifier module $h_S = h_T$ is kept in the target model adaptation for different domain-specific feature learning, and the parameters of the fault classifier are locked. To effectively train in the target domain, it is essential to fine-tune the feature extractor. It can extract the relevant domain-invariant features in the target domain and achieve a high-confidence output.

During the fine-tuning process, the predicted target samples will be mis-matched to other source hypothesis, which can be alleviated by self-training pseudo-label learning. The centroid Ξ_j for each predicted class j can be gotten by weighted clustering, which is shown in Eq. (12):

$$\Xi_j^{(0)} = \frac{\sum_T q_{pred,j} \cdot g_T}{\sum_T q_{pred,j}} \quad (12)$$

$$= \frac{\sum_T \Theta_{h,j}(G_g(x_T)) \cdot G_g(x_T)}{\sum_T \Theta_{h,j}(G_g(x_T))}$$

where $\Xi_j^{(0)}$ is the initial class-wise feature-level centroid to the j th faulty class, $q_{pred,j}$ is the j th channel of classifier's softmax output as shown in Eq. (2). $\Theta_{h,j}$ and G_g represent the mapping function of classifier and feature extractor as shown in Eqs. (1) to (2). x_T means the target data input.

After the centroids $\Xi_j^{(0)}$ calculation, the pseudo-labels $y_T^{(0)}$ can be gotten by the metric-based categories assignment. We use the cosine distance to describe the differences between samples, as shown below:

$$y_T^{(0)} = \underset{j}{\operatorname{argmin}} \operatorname{Dis}(G_g(x_T), \Xi_j^{(0)}) \quad (13)$$

$$= \underset{j}{\operatorname{argmin}} \left(1 - \frac{G_g(x_T) \cdot \Xi_j^{(0)}}{\|G_g(x_T)\| \cdot \|\Xi_j^{(0)}\|} \right)$$

where $y_T^{(0)}$ is the initial pseudo-labels of unknown-label target data. $\operatorname{Dis}(\cdot, \cdot)$ means the metric-based measure, which is cosine distance in this paper. And the minimization of related categories is assigned to the feature as labels.

Such a process of centroid computation and pseudo-label assignment is repeated several times as an iteration to achieve more accurate pseudo-labels. And the repeated works as shown in Eqs. (14) to (15):

$$\Xi_j^{(i)} = \frac{\sum_T \mathbf{1}(y_T^{(i-1)} = j) \cdot G_g(x_T)}{\sum_T \mathbf{1}(y_T^{(i-1)} = j)} \quad (14)$$

$$y_T^{(i)} = \underset{j}{\operatorname{argmin}} \left(1 - \frac{G_g(x_T) \cdot \Xi_j^{(i-1)}}{\|G_g(x_T)\| \cdot \|\Xi_j^{(i-1)}\|} \right) \quad (15)$$

where $\Xi_j^{(i)}$ is the centroid of the i th iterative time and j th predicted class, $y_T^{(i)}$ is the i th iterative updated pseudo-labels. $\mathbf{1}(\cdot)$ is an indicator that equals 1 when the argument is true.

With the continuous iteration of pseudo labels, the proposed model is gradually self-training, and the loss function of this mechanism is as follows:

$$L_T^{(1)} = -\mathbb{E}_T \left(\sum_{j=1}^C \mathbf{1}(y_T^{(end)} = j) \log \left(\frac{\exp(\Theta_{h,i}(g_S))}{\sum_{j=1}^C \exp(\Theta_{h,j}(g_S))} \right) \right) \quad (16)$$

where $\mathbb{E}_T$ is the mean output of each batch-size target samples, $y_T^{(end)}$ is the final pseudo-labels.

2.3.2. Entropy loss constraint

An ideal output of the softmax layer of the predictor is characterized by high-information and low-entropy. The original source model can achieve accurate classifications on data with the same I.I.D. condition, but will have reduced discriminative power and its associated classification results will have more uncertainty in the target classification task. Therefore, if there are no perturbations and domain shifts, the original source model can achieve accurate classifications and its output should be close to the one-hot encoding.

In contrast, the occurrence of domain shift becomes challenging to differentiate between related categories of samples because the predictive outputs for several faulty classes appear quite similar. From the classification perspective, the distance between the classification interface and the feature points is too close, as shown in Fig. 6, which can be seen as a poor generalization ability and overfitting.

Based on this principle, the self-entropy loss $L_T^{(2)}$ is designed as an optimization objective to increase the distance between the feature points and the classification interface. Such a design of optimization objective helps to enhance the confidence level of prediction results, as shown below:

$$\begin{aligned} L_T^{(2)} &= -\mathbb{E}_T \left(\sum_{i=1}^C q_{pred} \log q_{pred} \right) \\ &= -\mathbb{E}_T \left(\sum_{i=1}^C \frac{\exp(\Theta_{h,i}(g_T))}{\sum_{j=1}^C \exp(\Theta_{h,j}(g_T))} \log \frac{\exp(\Theta_{h,i}(g_T))}{\sum_{j=1}^C \exp(\Theta_{h,j}(g_T))} \right) \end{aligned} \quad (17)$$

To make the outputs of unknown target data similar to a one-hot vector, the one-hot encoding can be limited by the entropy-based loss. To lower information uncertainty and enhance the accuracy of predicting classification objectives, the information maximization (IM) loss is proposed here.

However, the high-performance entropy loss optimization objective often leads the output of neural networks to the other extreme, that collapsing into the same one-hot vector. To reduce this negative impact and ensure the diversity of outputs, an additional diversity loss $L_T^{(3)}$ is used as a regularization term, and the detailed form is shown below:

$$\begin{aligned} L_T^{(3)} &= \sum_{i=1}^C \mathbb{E}_T(q_{pred}) \log \mathbb{E}_T(q_{pred}) \\ &= D_{KL} \left(\mathbb{E}_T(q_{pred}) \parallel \frac{1}{C} \mathbf{1}_C \right) - \log(C) \\ &= D_{KL} \left(\frac{\exp(\Theta_{h,i}(g_S))}{\sum_{j=1}^C \exp(\Theta_{h,j}(g_S))} \parallel \frac{1}{C} \mathbf{1}_C \right) - \log(C) \end{aligned} \quad (18)$$

where $\mathbb{E}_T(q_{pred})$ is the mathematical expectations of the batch-size target domain. $D_{KL}(p \parallel q)$ is the Kullback-Leibler divergence, which is equal to $\sum p \log(p/q)$. $\mathbf{1}_C$ is a C-dimensional vector, and each value is all one.

The optimization objective can be formed as L_T^{all} , includes the self-training and entropy loss constraints, as shown in Eq. (19):

$$L_T^{all} = L_T^{(1)} + \lambda_2 L_T^{(2)} + \lambda_3 L_T^{(3)} \quad (19)$$

where λ_2 and λ_3 are the balancing hyper-parameter to control the impact level of entropy loss term. And the update of model parameters

are shown below:

$$\begin{aligned} \theta_g^t &= \theta_g^{t-1} - \epsilon_T \cdot \frac{\partial L_T^{all}}{\partial \theta_g} \\ &= \theta_g^{t-1} - \epsilon_T \cdot \frac{\partial (L_T^{(1)} + \lambda_2 L_T^{(2)} + \lambda_3 L_T^{(3)})}{\partial \theta_g} \end{aligned} \quad (20)$$

where ϵ_T is the learning rate, θ_g^t represents the updated feature extractor in the target at t epoch, and θ_h is unchanged during the target domain adaptation.

2.4. Algorithm implementation

The proposed G-SFDA method consists of two components: generalized source model training and target model adaptation, which are prerequisites for addressing domain shift and source privacy protection in fault classification tasks. The proposed method applies to scenarios where the data distribution in the source and target domains differs, and only model parameters (and not data) can be shared, as shown in Fig. 7. To better represent the training and test procedures of the proposed framework, we present a pseudo-code in Algorithm 1 to illustrate the necessary steps.

Algorithm 1: Procedure of proposed G-SFDA method for fault diagnosis tasks

Input: Training source datasets $\{X_S, Y_S\}$, training target datasets $\{X_{Ta}, Y_{Ta}\}$, Test datasets $\{X_{Te}, Y_{Te}\}$

Output: Detailed model parameters of G-SFDA model, predicted results $\hat{Y}_{Te}$

```

1 Data acquisition and preprocessing;
2 Model initialization and preparation;
// Training steps on source model
3 while Epochnow ≤ Epochmax do
4   Forward calculate  $\{X_S, Y_S\}$  by Eqs.(1) to (2);
5   Calculate the loss value  $L_S^{(1)}$  by Eq. (3);
6   Get the new pairs  $g'_i, g'_j$  based on the manifold Mixup augmentation by Eqs. (4) to (6);
7   Get the manifold Mixup loss value  $L_S^{(2)}$  by Eqs. (7) to (8);
8   Calculate the all loss value  $L_S^{all}$  by Eq. (9) and update the source model parameters  $\theta_g$  and  $\theta_h$  by Eqs. (10) to (11);
9   Epochnow ← Epochnow + 1;
10 end
// Training steps on target model
11 while Epochnow ≤ Epochmax do
12   Get the pseudo-labels  $q_{pred}$  based on target datasets  $\{X_{Ta}\}$ ;
13   Calculate the centroids  $\mathcal{E}_j$  of each pseudo-labels  $q_{pred,j}$  and get the metric-based assigned categories by Eqs. (12) to (15);
14   Calculate the self-training loss value  $L_T^{(1)}$  by Eq. (16);
15   Get the self-entropy loss  $L_T^{(2)}$  as a design of optimization objective by Eq. (17);
16   Get an additional diversity loss  $L_T^{(3)}$  as a regularization term by Eq. (18);
17   Calculate the all loss value  $L_T^{all}$  by Eq. (19) and update the target model parameters  $\theta_g$  by Eq. (20);
18   Epochnow ← Epochnow + 1;
19 end
// Online test steps
20 Do fault classification on the test sets  $\{X_{Te}\}$  and get evaluated indicators of results;
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Steps 1 and 2 depict the principal data preprocessing and model initialization tasks. Steps 3 to 10 represent the generalized source model training process, which relies on known source datasets X_S and Y_S . Steps 11 to 19 show the target model adaptation process on target

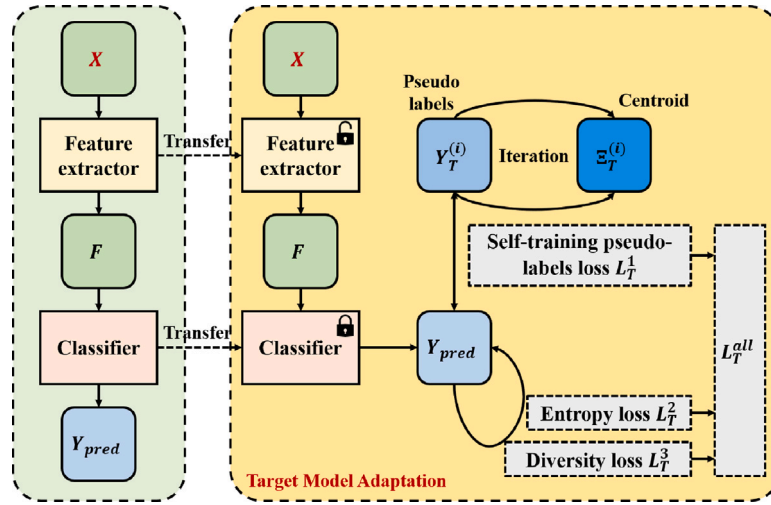


Fig. 7. Diagram of target model adaptation.

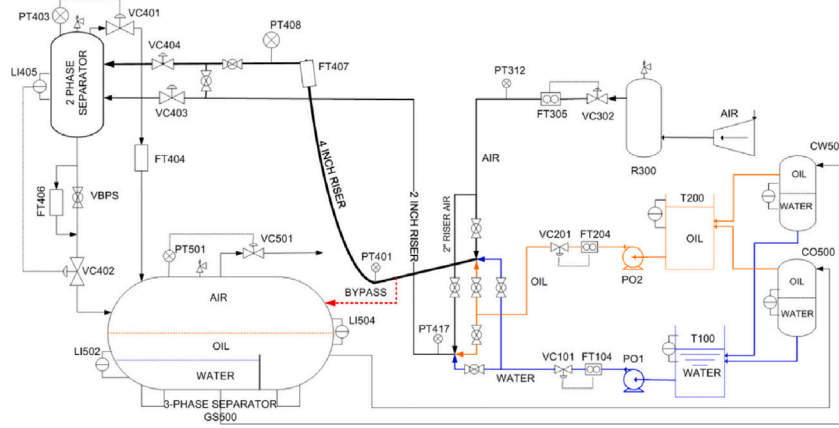


Fig. 8. The diagram of three-phase flow industrial process.

datasets X_{Ta} with unknown labels. Finally, step 20 shows the online testing process on test dataset X_{Te} .

3. Case study 1: Three-phase flow equipment fault diagnosis

3.1. Dataset description and experiment setup

In industrial production, a three-phase flow process measures and controls a pressure system's water, oil, and air flow rate. It is a complex industrial experiment that involves multiple operating conditions [35]. During the process, the supervisory, control, and data acquisition (SCADA) system can retrieve and collect time-stamped data from samples, and all data is collected, including 24 different main signals for health monitoring that reflect internal and external situations.

The diagram of the three-phase flow device is shown in Fig. 8. In this paper, we investigate four types of faults for fault diagnostics experiments. A detailed introduction of the faulty types is given in Table 1. Water flow and air flow are the two key process inputs that vary across operating conditions. We divide the datasets into two different operating conditions based on the detailed values of the water flow. An overview of the working conditions for the three-phase flow process is given in Table 2. Two tasks are used as validation, including the domains $A \rightarrow B$ and $B \rightarrow A$. There are a total of 4000 source samples (X_{Tr} , Y_{Tr}) with 1000 samples for each class during source model generation, and 400 target samples (X_{Ta}) with 100 samples for each class during target model adaptation, and 4000 target samples (X_{Te}) as the test set.

Table 1

Fault description of three-phase flow process.

Faulty types	0	1	2	3
Description	Normal condition	Air line blockage	Water line blockage	Top separator input blockage

Table 2

Overview of the working conditions of three-phase flow process.

Domain	Conditions	Class 1	Class 2	Class 3	Class 4
A	Water flow (kg/s)	2	2	2	2
	Air flow (m ³ /s)	0.0278	0.0417	0.0278	0.0278
	No. of samples	1000	1000	1000	1000
B	Water flow (kg/s)	3.5	3.5	3.5	3.5
	Air flow (m ³ /s)	0.0208	0.0208	0.0417	0.0208
	No. of samples	1000	1000	1000	1000

3.2. Comparison methods and experiment setup

We introduce traditional methods, including the SVM, DT, RF, KNN, and deep neural network (DNN), for a complete comparison. For these non-adaptation machine learning algorithms, the training set is only in source data, and the classification task is done in target data.

To validate the performance of the proposed SFDA method, we also introduce some domain adaptation methods for comparison. We

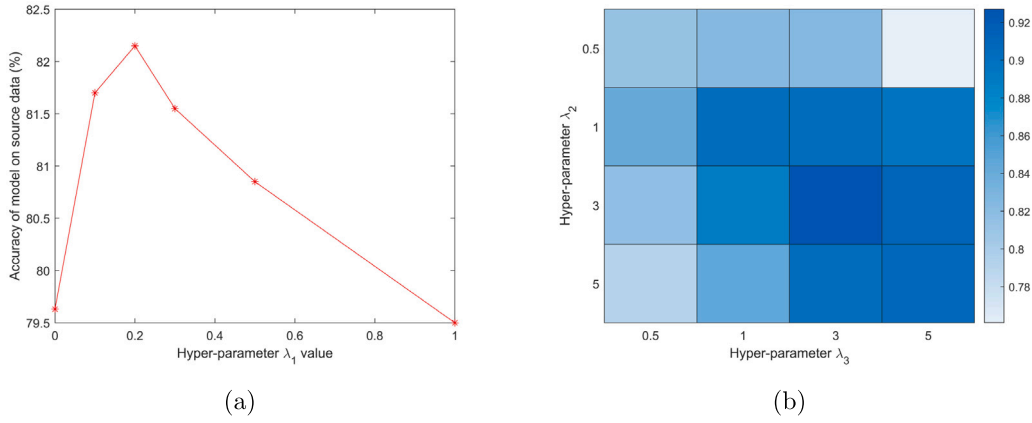


Fig. 9. Adjusted experiments on trade-off parameters λ_1 (a), λ_2 , λ_3 (b).

apply some hot deep learning-based domain adaptation methods for comparison, and the detailed descriptions are as follows:

- **Domain Adaptation Networks (DAN)** [36]: The deep neural network has a feature representation learning process that includes a domain adaptation module, which uses the maximum mean discrepancy to extract domain-invariant features across all domains.
- **Deep CORal (DCO)** [37]: Similar to DAN, Coral serves as a different metric to find domain-invariant features.
- **Deep Subdomain Adaptation Networks (DSAN)** [38]: A local maximum mean discrepancy is used to constrain the transfer loss in the classification task.
- **Source HypOthesis Transfer (SHOT)** [23]: A mainstream source-free domain adaptation method to realize the transfer task in classification.
- **Source-free domain adaptation with confidence-based filter (SFDA-CF)** [28]: Based on the SHOT model, improve the confidence of pseudo labels by filters.
- **Source-free domain adaptation with nuclear norm maximization (SFDA-NNM)** [27]: The nuclear norm is applied to express the confidence of labels.

For these domain adaptation-based methods, the training set consists of known labels from the source and unknown labels from the target. DAN, DCO, and DSAN are used as classic deep domain adaptation methods, and SHOT, SFDA-CF, and SFDA-NNM are applied as novel source-free domain adaptation methods for comparison. The metric indicators accuracy and F1-score are applied here to compare the classified performance.

3.3. Implementation details in model training

There are totally 24-dimensional features from sensors of the three-phase flow industrial system, which are applied for model training. Meanwhile, to eliminate other unknown influence factors, max-min normalization and outlier reduction are applied through preprocessing efforts. During the training process, a dynamic learning rate with weight decay is used, which is equal to $lr = lr_{initial} \cdot (1 + 10 \cdot k / \text{maxepoch})^{-0.75}$, where k denotes the current epoch. To ensure the consistency of all neural networks, we adopt the same hyper-parameters for these deep learning networks. We define the value of batch-size as 128 and the initial learning rate $lr_{initial}$ as 0.005 for all proposed and compared methods.

There are multiple loss terms with three trade-off coefficient parameter λ_1 , λ_2 , and λ_3 , which should be discussed in depth. The adjusted experiments on the trade-off weights are shown in Fig. 9. In Fig. 9(a), the X-axis represents the value of the trade-off parameter λ_1 , and the Y-axis shows the classification accuracy of the source model on the

Table 3

Detailed information of all compared approaches.

Approaches	Detailed information
SVM	Kernel: Gaussian
DT	Min samples split = 2
KNN	K = 1
RF	Estimators = 100 based on DT
DNN, G-DNN, DAN	Feature extractor:
DCO, DSAN, SHOT	Dense(100)-Dense(100)-Dense(100)
SFDA-CF, SFDA-NNM, SFDA-Ours	Fault classifier:
	Dense(100)-Dense(50)-Dense(4)

target data. λ_1 represents the influence of the manifold Mixup module on the source model. In Fig. 9(b), the horizontal axis represents the value of the trade-off parameter λ_2 , the vertical axis shows the value of the trade-off parameter λ_3 , and the color of the corresponding square represents the classification accuracy of the adapted model on the target data when selecting the current trade-off parameters. The darker the color, the higher the classification accuracy. λ_2 and λ_3 show the effect of the entropy loss term on the target model adaptation. Based on the results of adjusted experiments, we select a trade-off weight of 0.2 for λ_1 , and 3 for both λ_2 and λ_3 .

Meanwhile, the number of training epochs is set to 50 during source model generation and target model adaptation respectively. We select information about the model structure as shown in Table 3, including the detailed parameter values setting of compared traditional and other SOTA methods.

3.4. Results, analysis, and discussion

Experiments are conducted on cross-domain industrial system fault classification tasks to validate the proposed source-free domain adaptation method on industrial system fault diagnosis. This sub-section is used to show the results, analysis, and discussion.

3.4.1. Results

Table 4 shows the compared results of various fault diagnosis approaches. First, traditional methods including the SVM, DT, RF, KNN can achieve the accuracy between 50% and 75%, which is unsatisfactory. They are not able to perform well for multi-class classification tasks. Second, DNN based methods can improve the accuracy from 75% to 80%. They can achieve almost 100% accuracy on the source training set, but only 80% accuracy on the target task. The primary reason is that they concentrate too much on learning from one specific source, causing them to struggle with applying that knowledge to other similar domains. To deal with this issue, the proposed generalized DNN method in Section II-B is used here, with enhanced accuracy and F1-score

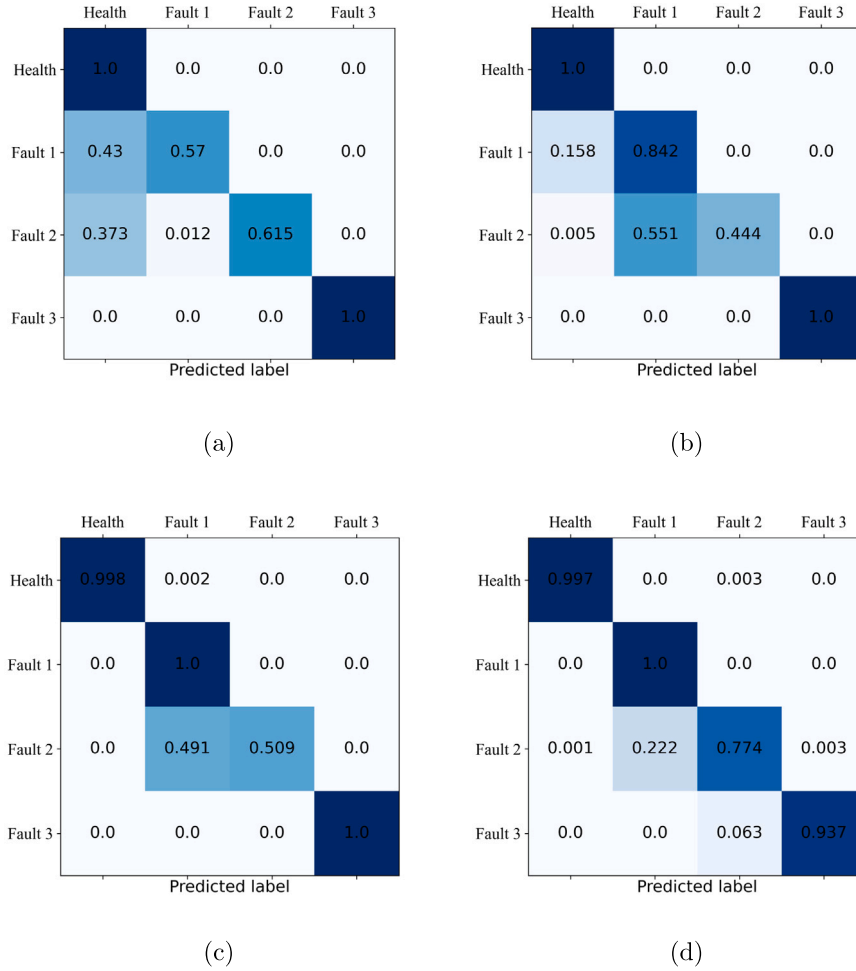


Fig. 10. Confusion matrix of multiple deep learning-based approaches for target domain data. (a) DNN. (b) G-DNN. (c) SHOT. (d) SFDA-Ours.

values. Then, the proposed SFDA is used to achieve semi-supervised self-learning target model adaptation, and the accuracy is around 92%, which outperforms of all the compared approaches.

The confusion matrix of the deep learning-based methods is shown in Fig. 10 to provide a more analytical view of the results. They are obtained from the domain $A \rightarrow B$ task. The classified results of the model can be seen on the X -axis and the actual labels on Y -axis. A higher-precision prediction implies a darker color of the diagonal. DNN-based methods show better predictions on healthy state and faulty class 3, but less performance on faulty classes 1 and 2. After the target model adaptation, the model can get high-performance classification on healthy state and faulty classes 1, 3. Therefore, the domain adaptation based methods can improve the results better. To deal with the source-free landscape, the proposed SFDA method realize the adaptation task well and get better performance from the confusion matrix.

3.4.2. Training process analysis

We delve into the training process for both two domains. From Fig. 11, it is already evident that the accuracy increases with the source training process and target model adaptation. For conventional DNNs, the accuracy is close to 100% on the source training set. Although such near-perfect accuracy is expected for the model, the presence of domain shift often leads to overfitting, which achieves worse results on the target data. The proposed manifold Mixup-based generalization module can sacrifice partial accuracy during training in the source domain and contains around 98% accuracy, but can achieve better

results and enhance generalization on other datasets. The accuracy of model adaptation in the target domain can be significantly improved with the designed loss function, as shown in Fig. 11(b). The result is a nearly 10% increase, maintaining the accuracy at around 92%.

3.4.3. Ablation study analysis

An ablation study is conducted here to investigate each component of the loss function L_S , $L_T^{(1)}$, $L_T^{(2)}$, $L_T^{(3)}$, and the entropy loss couple $L_T^{(2)} & L_T^{(3)}$. Table 5 shows the detailed results of the ablation study. The loss $L_T^{(1)}$ and $L_T^{(2)} & L_T^{(3)}$ have the more important impact on prediction accuracy. The model performance is decreasing by 7% to 9% in the absence of these losses. In target domain adaptation, the absence of either term leads to adverse transfer effects, so neither term is indispensable. The absence of $L_T^{(2)}$ or $L_T^{(3)}$ results in a reduction in the model's accuracy by about 5%, which is higher than the accuracy of the original DNN model with poor improvement. Meanwhile, the proposed Manifold Mixup loss term can improve accuracy by 3%, which helps the model increase its upper performance. The results show that each designed loss function significantly impacts the model's prediction accuracy. Therefore, these losses are essential for domain adaptation tasks under source-free cross-domain conditions.

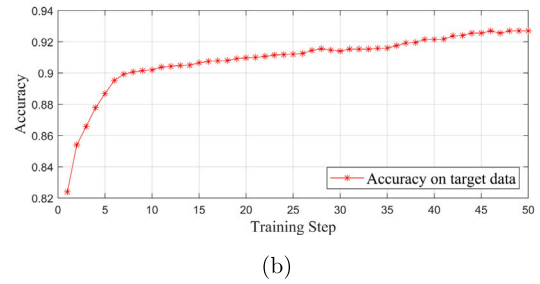
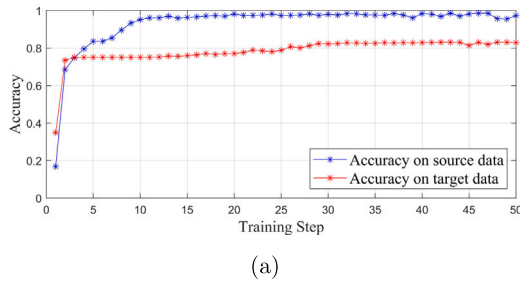
3.4.4. Discussion

The accuracy of our proposed SFDA methods has surpassed all the compared approaches. Time complexity analysis of algorithms is crucial as it indicates whether they can be used in the real world. We analyze

Table 4

Performance comparison of different fault diagnosis approaches in three-phase flow process.

Method	Domain A $\rightarrow$ B		Domain B $\rightarrow$ A		Average		DA	SF
	Accuracy	F1-score	Accuracy	F1-score	Accuracy	F1-score		
SVM	0.7525	0.7541	0.7187	0.7102	0.7356	0.7322	✗	✗
DT	0.5367	0.4074	0.5890	0.5818	0.5629	0.4946	✗	✗
RF	0.6730	0.6056	0.6252	0.6048	0.6491	0.6052	✗	✗
KNN	0.7232	0.7139	0.7090	0.7015	0.7161	0.7077	✗	✗
DNN	0.7963	0.7989	0.8083	0.8076	0.8023	0.8033	✗	✗
G-DNN	0.8215	0.8108	0.8295	0.8295	0.8255	0.8202	✗	✗
DAN	0.8505	0.8490	0.8390	0.8204	0.8448	0.8347	✓	✗
DCO	0.8730	0.8622	0.8215	0.8215	0.8473	0.8419	✓	✗
DSAN	0.8801	0.8728	0.8870	0.8807	0.8836	0.8768	✓	✗
SHOT	0.8768	0.8690	0.8738	0.8769	0.8753	0.8730	✓	✓
SFDA-CF	0.9035	0.9012	0.8962	0.8981	0.8999	0.8997	✓	✓
SFDA-NNM	0.8905	0.8897	0.8938	0.8886	0.8922	0.8892	✓	✓
SFDA-Ours	0.9270	0.9263	0.9170	0.9179	0.9220	0.9221	✓	✓

**Fig. 11.** Accuracy increasing trend under training steps of source model generation (a) and target model adaptation (b) for the diagnostic task from domain A to B.**Table 5**

Ablation study of the proposed method.

Method	Domain A $\rightarrow$ B		Description
	Accuracy	F1-score	
SFDA-Ours	0.9270	0.9263	
w/o $L_S^{(2)}$	0.8990 $\downarrow$ 3.02%	0.8929 $\downarrow$ 3.61%	Mix-up loss
w/o $L_T^{(1)}$	0.8515 $\downarrow$ 8.14%	0.8537 $\downarrow$ 7.84%	Self-training constraint
w/o $L_T^{(2)}$	0.8780 $\downarrow$ 5.29%	0.8734 $\downarrow$ 5.71%	Self-entropy loss
w/o $L_T^{(3)}$	0.8768 $\downarrow$ 5.42%	0.8720 $\downarrow$ 5.86%	Diversity loss
w/o $L_T^{(2)} \& L_T^{(3)}$	0.8590 $\downarrow$ 7.34%	0.8574 $\downarrow$ 7.44%	All entropy loss

Table 6

A comparison of time complexity analysis for fault diagnosis methods.

Methods	DNN	SHOT	SFDA-Ours
Time (s)	615.28	1053.56	1064.68

the time complexity comparison of traditional DNN, SHOT, and SFDA in the same environment, as shown in Table 6. SHOT and G-SFDA algorithms have higher complexity than the classic DNN method due to the target model adaptation process. Besides, there is almost identical time complexity between the original SHOT and our proposed SFDA method, which shows that the internal manifold Mixup module has less impact on the complexity.

To validate the generalization of the SFDA framework with different neural network structures, we test additional experiments on various datasets here. A comparison of different neural network (fault classifier) structures for fault diagnosis task A to B is shown in Table 7. The model structure can impact the fault classification accuracy. Besides, the proposed SFDA can achieve around 92% accuracy, higher than the classic SHOT and DNN methods. The results can validate the algorithm's generalization.

G-SFDA can reduce the dependency on source data, protecting privacy and preventing data transmission. For the same three-phase flow fault diagnostic tasks, the proposed SFDA can only require the

Table 7

Accuracy comparison of different neural network structures for fault diagnosis task A to B.

Structure	DNN	G-DNN	SHOT	SFDA-Ours
50-20-4	0.7690	0.8183	0.8892	0.9172
100-50-4	0.7963	0.8215	0.8768	0.9270
100-100-4	0.7933	0.8380	0.8950	0.9153

Table 8

A comparison of time complexity analysis for fault diagnosis methods.

Domain	Load (HP)	Healthy samples	Faulty samples
C	1	945	1189
D	2	947	1187
E	3	948	1192

consumption of the model transmission instead of data transmission. Therefore, it has efficient applications in industrial systems.

4. Case study 2: CWRU bearing diagnostics

We apply the bearing datasets from Case Western Reserve University (CWRU) to verify the performance of the proposed approach. The electrical discharge machining (EDM) technique is applied for bearings to set faults. To better represent the bearings' information, we extract 44 critical features from time and frequency domains, the same as the Ref. [13].

The faulty classes can be divided into the ball fault (BF), inner race fault (IRF), and outer race fault (ORF). We also classify the ORF into three faulty classes based on the damaged diameter, and we can get five fault classes and one health class. We select three operating conditions with different loads to represent the domain shift phenomenon, and the detailed information is shown in Table 8.

On this basis, we conduct experiments on six tasks on this dataset and set the specific task conditions. For these tasks, the performance comparison in bearing datasets is shown in Table 9. The results show

Table 9
Performance comparison in CWRU bearing datasets (Accuracy).

Method	C → D	D → C	C → E	E → C	D → E	E → D	Average
SVC	0.9311	0.9808	0.9575	0.9091	0.9722	0.8918	0.9404
DT	0.9642	0.9859	0.9346	0.9360	0.9589	0.9349	0.9524
RF	0.9699	0.9895	0.9355	0.9630	0.9612	0.9429	0.9603
KNN	0.9681	0.9836	0.9332	0.8903	0.9694	0.9072	0.9420
DNN	0.9721	0.9905	0.9456	0.9776	0.9879	0.9789	0.9754
G-DNN	0.9794	0.9937	0.9673	0.9811	0.9881	0.9884	0.9830
DAN	0.9752	0.9920	0.9723	0.9869	0.9903	0.9870	0.9840
DCO	0.9723	0.9901	0.9546	0.9770	0.9874	0.9812	0.9771
DSAN	0.9805	0.9948	0.9861	0.9890	0.9960	0.9913	0.9896
SHOT	0.9974	0.9948	0.9705	0.9902	0.9888	0.9934	0.9892
SFDA-NNM	0.9912	0.9916	0.9859	0.9923	0.9986	0.9944	0.9923
SFDA-CF	0.9993	0.9930	0.9895	0.9916	0.9895	0.9965	0.9932
SFDA-Ours	0.9986	0.9991	0.9909	0.9995	0.9958	0.9979	0.9970

Table 10
Performance comparison in SEU bearing datasets (Accuracy (%)).

Method	DNN	G-DNN	DAN	DSAN	SHOT	SFDA-Ours
F → G	81.95	84.97	95.49	96.67	96.80	98.00
G → F	82.51	84.89	96.93	97.80	98.20	98.60
Average	82.23	84.93	96.21	97.24	97.50	98.30

that SVC, DT, RF, and KNN models can reach around 95%. As a neural network approach, DNN and G-DNN can enhance the accuracy of fault diagnosis tasks, and domain adaptation methods can improve the accuracy from 97% to 98%. The source-free domain adaptation framework can achieve 99% accuracy, and our proposed G-SFDA approach can get the best performance.

5. Case study 3: SEU bearing diagnostics

We apply the real-world bearing data from Southeast University (SEU) to verify the algorithms' performance [39]. There is one healthy class and four faulty classes for the state of bearings, and the dataset consists of 1000 samples for each category. For this bearing dataset, eight-channel signals are extracted, and each signal corresponds to 6 key features, including the mean, maximum, minimum, standard deviation, kurtosis, and zero-crossing rate, with 48 features as model inputs. The bearings work on two working conditions due to different loads of the speed systems, including 20HZ-0V and 30HZ-2V. Therefore, we set the domains as F and G and conduct the transfer tasks F → G and G → F.

We select the main neural network-based algorithms for these tasks and show the performance comparison in Table 10. The results show that there is a better accuracy for transfer methods compared with conventional neural networks. Domain shift issues have an obvious negative effect on the conventional neural networks in this case, and transfer learning can improve classification performance by considering distribution differences. Meanwhile, compared to traditional domain adaptation methods, our proposed approach can achieve 98% accuracy and get the best performance among these methods.

6. Conclusion

To tackle the domain shift under cross-domain conditions, we proposed a novel source-free domain adaptation with manifold Mixup data augmentation for fault diagnosis. The manifold Mixup is proposed as a new module to get a generalized model during the source model generation to tackle the overfitting issue. The self-training pseudo-labels cross-entropy and self-information entropy losses are used to realize the model adaptation. Compared with conventional domain adaptation methods, the proposed model can relieve the dependence on source data during adaptation and rely only on known trained source models in data privacy protection and big data flow transmission. To

validate the performance of our proposed approach, we utilize real-world industrial process datasets. Our approach has outperformed both traditional and SOTA methods. In future work, we will extend the high-performance domain adaptation under source-free and open-set fault diagnosis.

CRedit authorship contribution statement

Jilun Tian: Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Jiushi Zhang:** Methodology, Writing – review & editing. **Yuchen Jiang:** Writing – review & editing, Conceptualization. **Shimeng Wu:** Conceptualization, Methodology. **Hao Luo:** Writing – review & editing, Supervision, Funding acquisition. **Shen Yin:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are openly available.

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